

increase in labor productivity by 4% when instrumenting for AI adoption – a plausible figure that aligns with the more moderate effects found in the literature and discussed in Section 2. For the firm-specific controls, the results remain consistent with those in column (2). An exception is investment, which is now also statistically significant and positive (i.e. firms that invest more are also more productive).

Table 2: Effects of AI adoption on labor productivity

	(1)	(2)	(3)	(4)
	LP (log)	LP (log)	LP (log)	LP (log)
	No controls	With controls	No controls	With controls
AI adoption (actual)	0.162*** (0.011)	0.028*** (0.011)		
AI adoption (US-based)			0.156*** (0.03)	0.041*** (0.019)
Investment (lagged)		0.009* (0.005)		0.009*** (0.003)
Profitability (lagged)		0.590*** (0.038)		0.618*** (0.019)
Financial leverage (lagged)		-0.178*** (0.025)		-0.151*** (0.012)
Total assets (log, lagged)		0.386*** (0.006)		0.386*** (0.002)
Size FE	Yes	Yes	Yes	Yes
Age FE	Yes	Yes	Yes	Yes
Country-sector-wave FE	Yes	Yes	Yes	Yes
Observations	53730	35648	44844	30228
R-squared	0.373	0.541	0.380	0.551
Adjusted R-squared	0.352	0.519	0.355	0.526

Note: The dependent variable is labor productivity, calculated as the log of turnover per employee, and is derived from EIBIS. AI adoption (actual) is the EIBIS-reported AI implementation status. AI adoption (US-matched) is the AI implementation status derived from similar firms in the US. Investment is expressed as the annual change in total fixed assets. Profitability is the ratio of EBIT to total assets. Financial leverage is the ratio of loans and long-term debt to total assets. All control variables come from ORBIS and are lagged by 1 year. Standard errors clustered at the country-sector-wave level in columns (1) and (2), and bootstrapped in columns (3) and (4) are reported in parentheses. *, **, *** correspond to 0.1, 0.05 and 0.01 significance levels, respectively.